

SRIDHARAN KANNAN

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Education:

Amrita Vishwa Vidyapeetham, Coimbatore

July 2024 - Present

Bachelor of Technology, Computer Science & Engineering | CGPA 8.63/10 (up to 4th semester)

Relevant Coursework: Data Structures and Algorithm, Object Oriented Programming, Operating Systems, Computer Organization and Architecture, Discrete Mathematics, Linear Algebra, Probability and Random Processes.

Projects:

CppKV

Individual Project

[GitHub](#); Tech Stack: C++ 20, Windows IOCP, RESP

- Built a Redis compatible in-memory key-value store in C++ 20, achieving ~59,700 req/sec and outperforming Memurai (native Windows Redis) by 2.4x across SET, GET, INCR, LPUSH, RPUSH, LPOP, RPOP, LRange(100-600 elements) benchmarks.
- Designed a sharded execution model with 16 independent shard owner threads and MPSC queues for lock-free data access, no mutex on hot path data.
- Implemented crash resilient persistence via a Write-Ahead Log(AOF) and a binary RDB snapshot, with startup replay to recover in case of failures.
- Built a primary-replica replication using PSYNC handshake, full resync with RDB transfer, and a per replica feeder thread streaming from an in-memory replication backlog.
- Implemented consistent hash ring clustering with virtual nodes and MOVED redirection, enabling key based routing across multi-node deployments.

HopOn

Team Project

[GitHub](#); [Live Demo](#); Tech Stack: JavaScript, Firebase Realtime Database, Leaflet.js, OpenStreetMap, OSRM

- Built a Progressive Web App (PWA) for real-time bus tracking, installable on mobile and desktop, with role-based interfaces for admins, drivers, and passengers.
- Used Firebase Realtime Database for live location sync across all clients, enabling drivers to broadcast position updates and passengers to see buses move in real time.
- Implemented road accurate ETA computation using the OSRM routing API, with Haversine distance as a fallback when the routing service is unavailable.
- Built an interactive route builder using Leaflet.js and OpenStreetMap, allowing admins to place and name stops visually on a live map.

Technical Skills:

Programming Languages: C, C++ 20 (STL), Python, JavaScript, SQL, Shell Scripting

Tools and Technologies: Git, Firebase, MySQL, Leaflet.js

Concepts: Concurrent and Systems Programming, Network Programming, Distributed Systems

Competitive Programming: LeetCode Contest Rating: 1827